What Makes Grocery Customers Reorder?

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Summary

I analysed 3.4 million online grocery orders from 206,000 Instacart customers in MySQL to understand repeat buying: how often people come back, what they reorder and what they buy together. Repeat purchases dominate. 59% of all items ordered are reorders, and a small set of customers and products account for most of the activity.

- Weekly shoppers are 12% of customers but place 30% of all orders.
- 790 products (1.6% of those sold) make up half of all items ordered.
- Produce and dairy are 46% of all items, and both are reordered about two-thirds of the time.
- Items added to the cart first are reorders 68% of the time, compared with 46% for items added 20th or later.
- Clear product pairs exist. Sparkling water flavours are bought together 18 times more often than chance.

Data and approach

The data is the Instacart Market Basket Analysis dataset from Kaggle: six CSV files covering orders, the products in each order (33.8 million rows), products, aisles and departments. It has no prices or calendar dates, only the order sequence for each customer and the number of days since their previous order.

I loaded everything into MySQL 8.4. The products file escapes quotes in an unusual way, which made MySQL silently drop 333 products on the first load. I caught it by comparing row counts with the source file and fixed the load. Joining 33.8 million order lines to products for every question took several minutes per query. So I first aggregated the order lines into two summary tables, one row per product and one row per order, and ran most of the analysis on those. That brought the full analysis down to about 13 minutes on a laptop.

Findings

1. Customers shop on a routine

The most common gap between orders is exactly 7 days (10% of repeat orders), with smaller peaks at 14, 21 and 28 days. 11.5% of orders show a 30-day gap, but the dataset caps the value at 30, so these really mean 30 days or more. Grouping customers by their average gap between orders shows how concentrated the activity is:

Shopper type	Customers	Share of orders	Reorder rate
Weekly	12.4%	29.6%	58%
Fortnightly	34.0%	41.6%	51%
Every 3 weeks	30.8%	19.9%	40%
Monthly or less	22.8%	8.8%	32%

Customers who shop more often also reorder more of their basket, so frequency and habit seem to go together.

2. A small part of the catalogue does most of the volume

Of the 49,685 products that sold at least once, the top 790 make up 50% of all items ordered and the top 4,558 make up 80%. Bananas alone were ordered 491,000 times, with an 84.5% reorder rate. Among products with at least 5,000 orders, the highest reorder rates are almost all milk, at 83–86%.

3. Fresh food and dairy are habit purchases

Department	Share of items	Reorder rate
Produce	29.2%	65%
Dairy & eggs	16.7%	67%
Snacks	8.9%	57%
Beverages	8.3%	65%
Frozen	6.9%	54%
Pantry	5.8%	35%
Personal care	1.4%	32%

Pantry and personal care items have the lowest reorder rates, which makes sense since people stock up on them rather than buying them every week. Organic products are 31.5% of items and are reordered more often than non-organic ones (63.5% vs 56.9%).

4. The first items in the cart are the most predictable

The reorder rate falls steadily with cart position, from 68% for the first item to 55% for the tenth and 46% for the twentieth item onwards. It looks like customers add their regular items first and then browse for new things.

5. Products that are bought together

For the 100 most ordered products, I counted how often each pair appears in the same order and compared that with what you would expect by chance (lift).

Product A	Product B	Orders together	Lift
Sparkling Lemon Water	Lime Sparkling Water	8,874	17.9
Lime Sparkling Water	Sparkling Water Grapefruit	13,789	12.0
Jalapeno Peppers	Bunched Cilantro	6,034	9.6
Orange Bell Pepper	Red Peppers	5,529	7.7

Organic Garlic	Organic Ginger Root	11,236	6.1

Sparkling water is bought as a mix of flavours, and salsa and guacamole ingredients (jalapeño, cilantro, limes, red onion, tomatoes) keep appearing together.

6. When people order

Days 0 and 1 are the busiest, with 17–18% of orders each compared with about 13% on other days. The dataset does not say which weekday 0 is, but it is usually read as Sunday. Most orders are placed between 9am and 4pm, and very few come in overnight.

Recommendations

- Focus retention work on fortnightly and 3-week shoppers (65% of customers). Reminders timed to each customer's usual gap could move some of them to a weekly routine.
- Make reordering the default experience, with a "buy your usual" list on the home screen. Six in ten items are repeats, and the first items added are the most predictable.
- Offer bundles based on the strongest pairs, such as sparkling water variety packs, a salsa kit and a stir-fry kit with garlic, ginger and peppers.
- Prioritise stock availability for the roughly 800 products that make up half of all volume.

Limitations

Instacart only included customers with between 4 and 100 orders, so one-time buyers are missing and the true reorder rate across all customers would be lower. There are no prices, so the analysis covers volume, not revenue. The data is from the US in 2017, so the patterns will not carry over exactly to Indian quick-commerce apps.